cse 242 hw2

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1. (a)

$$\begin{aligned} J(w) &= \left|Xw - y\right|_2^2 + \lambda\left|w\right|_2^2 \\ J(w) &= (Xw-y)^T(Xw-y) + \lambda w^T w \\ J(w) &= w^T X^T X w - w^T X^T y - y^T X w - y^T y + \lambda w^T w \\ J(w) &= w^T X^T X w - 2y^T X w - y^T y + \lambda w^T w \\ \frac{\partial J(w)}{\partial w} &= 2X^T X w - 2y^T X + 2\lambda w = 0 \end{aligned} \tag{1}$$

then we see

$$\begin{aligned} 2X^T X w + 2\lambda w &= 2y^T X \\ (X^T X + \lambda I)w &= y^T X \end{aligned} \tag{2}$$

then we get the solution in the hw.

(b) as $\lambda \to 0$, $w_{ridge} -> (X^T X)^{-1} X^T y$. as $\lambda \to \infty$, $w_{ridge} -> 0$.
the former is when penalty goes to zero and the latter is when error dominates.

(c)

$$\begin{aligned} \left|\hat{X}w - \hat{y}\right| &= (\hat{X}w - \hat{y})^T(\hat{X}w - \hat{y}) \\ &= w^T \hat{X}^T \hat{X} w - \hat{y}^T \hat{X} w - w^T \hat{X}^T \hat{y} + \left|\hat{y}\right|^2 \\ &= w^T (X^T X + \sqrt{\lambda} X^T I_d + \sqrt{\lambda} I_d^T X + \lambda I) w \\ &\quad - \hat{y}^T \hat{X} w - w^T \hat{X}^T \hat{y} + \left|\hat{y}\right|^2 \end{aligned} \tag{3}$$

note $(\sqrt{\lambda} I w)^2 = \lambda \sum w_i^2$. factoring out w^T, excluding the $w = 0$ solution, the argmin when zero gives the original expression.

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(a)

$$\begin{aligned}(X^TX+\lambda I)^{-1} &= (Q\Lambda^2Q^T+\lambda I)^{-1} \\ &= (Q\Lambda Q^T+\lambda QIQ^T)^{-1} \\ &= (Q(\Lambda^2+\lambda I)Q^T)^{-1} \\ &= (Q^T)^{-1}((\Lambda^2+\lambda I))^{-1}Q^{-1} \\ &= Q((\Lambda^2+\lambda I))^{-1}Q^T\end{aligned} \tag{4}$$

λ inversely scales the eigenvalue

(b)

$$\kappa(X^TX) = \frac{\sigma_{max}^2}{\sigma_m^2 in} \tag{5}$$

since sigmas are positive, the inequality on the right holds.

the equality on the left is true because of part a. the eigenvalues are exactly $\Lambda + \lambda I$

we can also see why equality only hods when λ is zero, since every eigenvalue is different on the LHS when λ is nonzero.

(c) if there are zero eigenvalues, it means the matrix is non invertable. the smallest eigenvalues with regularization are λ and no eigenvalues are zero, which implies full rank, which implies invertability.

2. (a) for ols

$$(X^TX)^{-1}X^Ty = X^Ty \tag{6}$$

for ridge

$$(X^TX+\lambda I)^{-1}X^Ty = ((1+\lambda)I)^{-1}X^Ty = \frac{1}{1+\lambda}X^Ty \tag{7}$$

(b) using the hat symbol to denote estimate

$$\begin{aligned}E[\hat{w}_{ridge}] &= E\left[\frac{1}{1+\lambda}X^Ty\right] \\ &= \frac{1}{1+\lambda}X^TE[y]\end{aligned} \tag{8}$$

$$E[y] = E[Xw^* + \epsilon] = Xw^* + E[\epsilon] = Xw^*$$

then

$$E[\hat{w}_{ridge}] = \frac{w^*}{1+\lambda}$$

similarly

$$
\begin{aligned}
Var(\hat{w}_{ridge}) &= Var\left(\frac{1}{1+\lambda}X^T y\right) \\
&= \left(\frac{1}{1+\lambda}\right)^2 X^T Var(y)(X^T)^T \\
&= \frac{1}{(1+\lambda)^2} X^T Var(y) X \\
&= \frac{\sigma^2}{(1+\lambda)^2} X^T X
\end{aligned} \tag{9}
$$

(c)

$$
\begin{aligned}
\text{Bias}_j &= E[w_{ridge,j}] - w_j^* \\
&= \frac{w_j^*}{1+\lambda} - w_j^* \\
&= w_j^*\left(\frac{1}{1+\lambda} - 1\right) \\
&= w_j^*\left(\frac{1-(1+\lambda)}{1+\lambda}\right) \\
&= \frac{-\lambda w_j^*}{1+\lambda}
\end{aligned} \tag{10}
$$

therefore

$$MSE_j = \left(\frac{\lambda w_j^*}{1+\lambda}\right)^2 + \frac{\sigma^2}{(1+\lambda)^2}$$

to find optimal λ, w wolfram alpha

$$\frac{d}{d\lambda}MSE_j = \frac{2(\lambda(w_j^*)^2 - \sigma^2)}{(1+\lambda)^3}$$

set λ to $\frac{\sigma^2}{(w_j^*)^2}$ to make the numerator 0 and thus the whole term 0. OLS is bad when noise is high and good when noise is low. ridge is better when noise is high.

3. (a)

$$P(w|X,y) = \frac{P(y|X,w)P(w)}{P(y|X)}$$

$$\ln P(w|X,y) = \ln P(y|X,w) + \ln P(w) - \ln P(y|X)$$

both terms are normal distributions so we see

$$\ln P(y|X,w) = -\frac{1}{2\sigma^2}\|y - Xw\|_2^2 + C_1$$

$$\ln P(w) = -\frac{1}{2\tau^2}\|w\|_2^2 + C_2$$

note, after taking log the constant normalization terms for P become additive constants.

thus

$$\ln P(w|X, y) = -\frac{1}{2\sigma^2}\|y - Xw\|_2^2 - \frac{1}{2\tau^2}\|w\|_2^2 + C$$